

Can You Spot What's Real?

A Data Science Case Study on Detecting AI-Generated Images



You are a data scientist working on a media integrity team at a major social media platform. Lately, your team has been dealing with a growing issue: AI-generated images are spreading quickly, and it's getting harder to tell what's real and what isn't. Some of these images are harmless, but others can be used to spread misinformation or make fake events look real. The bigger problem is that newer AI tools can create images that look extremely convincing—sometimes even experts can't tell the difference.

Your team has been asked to look into whether data science can help.

You are given a large dataset of images that includes both real photos and AI-generated images from different tools. Your job is to explore whether machine learning models can actually tell them apart.

As you work through this, you'll think about questions like:

- Can a model reliably distinguish between real and AI-generated images?
- Can it go even further and figure out which AI model created an image?

You'll also run into some real challenges like dealing with imbalanced data, deciding how to train your model, and figuring out how reliable your results actually are.

By the end of this case study, you will create a deliverable in the form of a GitHub repository where you analyze the dataset, build and evaluate classification models, and explain what your results reveal about detecting AI-generated images. More importantly, you'll see how challenging this problem actually is, and why it's not as straightforward as it might seem.

So the question is: if even humans struggle to tell what's real... can a model do any better?